A STRUCTURED UPSKILLING ROADMAP

Building an AI-Ready Engineering Workforce

From a flagship AI-PDLC pilot hackathon to an institution-wide AI capability program

Prepared for the Chairman's Office | In collaboration with Heads of Department

WHY NOW

The AI market is scaling faster than the talent that can serve it

$2.59T

Global AI spending in 2026 — up 47% year-over-year (Gartner, May 2026)

$315B

India's tech industry revenue in FY26 — AI already a $10–12B revenue line (Nasscom)

920,000

India's AI-skilled professionals today — only 28% carry core AI depth (Nasscom, Jul 2026)

25–35%

India's AI market growth rate — still outpacing talent supply (Nasscom–BCG)

Even as overall IT hiring growth slows to roughly 2%, demand for AI, ML, data and cybersecurity talent is up 40–50% year-on-year — the market is rewarding readiness, not headcount.

Sources: Gartner (May 2026); Nasscom Annual Strategic Review 2026; Nasscom "Reorganise or Fall Behind" (Jul 2026)

THE SKILLS GAP IS ALREADY HERE

Employers are already spending heavily on AI upskilling — our graduates need to arrive ready

100,000+

employees each at TCS, Infosys & Wipro now on enterprise AI Copilot — up from 50K six months earlier (Microsoft, Jun 2026)

2,000,000+

professionals upskilled in AI industry-wide in FY26; 200–300K in advanced AI skills (Nasscom, 2026)

20,000,000

Indians Microsoft has pledged to AI-skill by 2030 — 5.6M already trained since Jan 2025

93%

of India's Gen Z already use AI in their day-to-day work (Deloitte, May 2026)

32%

of Gen Z professionals have completed formal AI training; 54% want more (Deloitte, 2026)

16%

of India's IT professionals are AI-skilled, even as AI job demand crosses 1M roles (MeitY/Nasscom)

Sources: Microsoft News (Jun 2026); Nasscom Annual Strategic Review 2026; Deloitte India Gen Z & Millennial Survey (May 2026); Nasscom / MeitY (2026)

DEFINING "AI-READY" — THE NPAI MODEL

AI skilling has three national tiers — and every discipline needs all three

FOR ALL

Basic Digital Literacy

Every student, every discipline. Comfortably use AI interfaces, understand responsible AI, and recognise its pitfalls — no exceptions.

FOR MANY

Digital Fluency

AI-empowered learners across every field — agriculture, healthcare, law, manufacturing, media — applying AI tools inside their own discipline.

FOR FEW

Digital Mastery

Deep technical builders: coding, ML, data analytics, cybersecurity, cloud — designing and shipping AI products and services.

AI-ready roles already exist in every sector — not only computer science

Agriculture
Precision Farming Technician

Healthcare
Healthcare AI Analyst

Legal
Explainable AI / Contract Review

Manufacturing
AI Process Engineer

Media
AI Content Creator

Governance
Responsible AI Officer

Source: National Programme on AI (NPAI) Skilling Framework, MSDE/NCVET (June 2023). Every AI course must devote 10%+ of its duration to Ethical AI.

NPAI SKILLING FRAMEWORK — SECTION 4.1

NPAI's own playbook for engineering colleges — and hackathons are the constant

ENGINEERING COLLEGES & STEM — MANY

Foundational AI, applied to every discipline

- Foundational courses: AI models, tools & frameworks, Responsible AI
- Cross-sector / multi-disciplinary AI application courses

Assessed via: knowledge tests, sandboxes, projects, internships, and **hackathons**

ENGINEERING COLLEGES & STEM — FEW

Advanced AI, building real products

- Advanced courses: Large Language Models, AI models & frameworks, Responsible AI
- Development of AI tools & products

Assessed via: badges, completions, and **hackathons / ideation**

Hackathons appear as NPAI's prescribed assessment method at every stage

Schools (Grades 6–12)	Non-IT / Non-Eng. Colleges	Engineering Colleges (Many)	Engineering Colleges (Few)	Industry
✓ Hackathon	✓ Hackathon	✓ Hackathon	✓ Hackathon	✓ Hackathon

Source: NPAI Skilling Framework §4.1, National AI Skilling Framework table (MSDE/NCVET, 2023). Hackathons are listed as the assessment mechanism for every segment shown above.

NPAI SKILLING FRAMEWORK — NEW JOB ROLES

AI is already creating new job roles — across every sector, not just IT

83M → 69M
Jobs projected displaced vs. created globally by 2028 (WEF)

22%
Projected churn in India's job market over the next five years

38%
Of that churn led by AI/ML roles — the single biggest driver (WEF)

New roles already appearing across NPAI's sector job-role tables

Prompt Engineer, Generative AI
IT-ITES · NSQF 4–7

Generative AI Engineer
IT-ITES · NSQF 6–7

AI Bias Engineer / Ethicist
IT-ITES · NSQF 5–7

Computer Vision Engineer
IT-ITES / Electronics · NSQF 5–7

Healthcare AI Analyst
Healthcare · NSQF 5–7

Precision Farming Manager
Agriculture · NSQF 5–7

AI Process Engineer
Iron & Steel · NSQF 6

Financial Data Analyst
BFSI · NSQF 6

AI Content Creator
Media & Entertainment · NSQF 4–6

Sports Data Analyst & Scientist
Sports · NSQF 6–7

Grid Engineer (AI-optimized)
Power · NSQF 5–7

Robotics Automation Lead
Electronics · NSQF 7

Every one of these roles is proven the same way — a working AI solution, built end-to-end. That's exactly what the AI-PDLC hackathon produces.

Source: NPAI Skilling Framework §9, Sector-Wise Analysis of Job Roles (MSDE/NCVET, 2023); WEF Future of Jobs Report (2023).

THE CHALLENGE — COGNITIVE SURRENDER

AI can give students the answer. It can't do the learning for them.

THE RISK

Cognitive surrender

Getting the right answer from an AI tool can feel like learning even when it isn't. MIT's Aug 2026 report on AI in education found students increasingly reach for AI at the first sign of difficulty — skipping the productive struggle that actually builds understanding. The same shift is quietly thinning out office hours, study groups, and the peer conversations where real learning happens.

The fix isn't less AI in the classroom. It's more structured, in-person work that AI can't shortcut.

WHAT FACULTY CAN DO

1 Structure group work with real check-ins

Weekly staff check-ins and deliverables that grade individual and team contribution separately, not just the final submission.

2 Put problem-solving back in the room

Lab-assistant or TA-guided sessions where students work through problems together, live — not alone with a chatbot.

3 Make feedback a conversation, not a comment

Discussions built around a shared class rubric, where students hear and respond to feedback in real time.

4 Grade participation, not just output

Facilitated in-class discussion where individual contribution is assessed directly — not inferred from a submitted file.

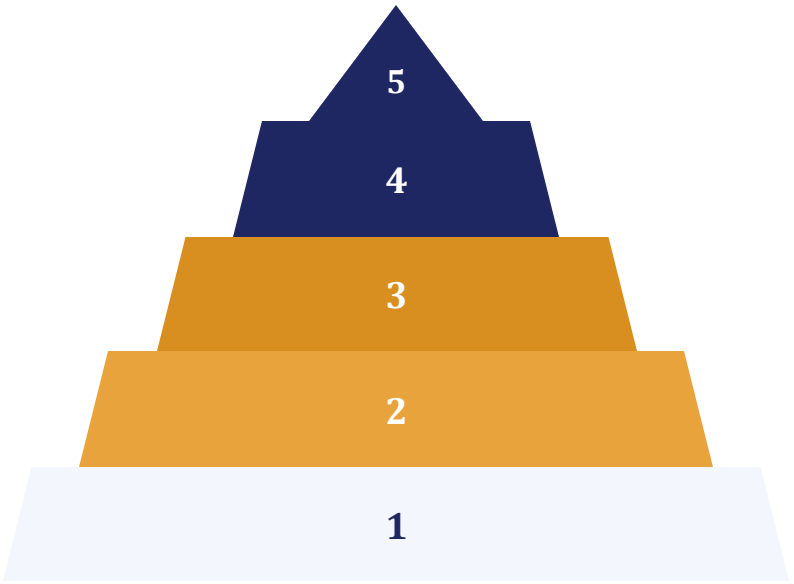
None of this is anti-AI. It's what MIT's own committee recommends for AI-era classrooms — real problems, worked through with real people, with AI as a tool inside that process, not a replacement for it.

Source: MIT Ad Hoc Committee on AI Use in Teaching, Learning and Research Training, Report (Aug 2026), Sections 2.4, 2.7, 3.1.4.

THE TALENT LADDER

Five tiers from AI literacy to AI mastery — mapped to real job roles

Research-grade mastery



Entry: every first-year student

AI Experts	*AI Researcher/Scientist, Generative AI Applied Researcher*	**NPAI: Few**
AI Builders / Developers	*ML/NLP Engineer, Generative AI Engineer, Computer Vision Engineer*	**NPAI: Few**
AI Integrators	*Solution Architect, Healthcare AI Analyst, Automotive Data Analyst*	**NPAI: Many**
AI Users	*AI-first Marketing Consultant, Precision Farming Technician*	**NPAI: Many**
AI Informed	*Every graduate — the AI Enabler / Digital Nagrik baseline*	**NPAI: All**

Our hackathon pilot targets AI Users → AI Integrators first

Source: NPAI Skilling Framework (MSDE/NCVET, 2023) job-role tables, cross-referenced with Nasscom's five-segment AI talent model.

OUR STRUCTURED APPROACH

A three-phase roadmap: Pilot, Scale, Institutionalize

1

Pilot

One department, one AI-PDLC hackathon

Prove the model with a 1.5-day flagship event: 6–12 teams, requirements to working demo.

2

Scale

Every department, every semester

Roll the hackathon format out department-by-department, including non-CS disciplines; embed foundational AI modules into coursework.

3

Institutionalize

Curriculum, faculty, and industry, permanently AI-aligned

Model curricula, Train-the-Trainer academies, Data & AI labs, and industry MoUs make this business-as-usual.

PHASE 1 · PILOT

The AI-PDLC Mini Hackathon: requirements to demo in 1.5 days

Unlike a conventional coding hackathon, teams don't jump to code. They live the full AI Product Development Lifecycle — market research, product definition, backlog, and only then implementation — using AI assistants (Claude, ChatGPT, Gemini) as collaborators at every stage.

- Explain the AI-PDLC stages and why requirements precede code
- Use AI to draft, critique and refine an MRD and PRD
- Decompose a PRD into epics and INVEST-quality user stories
- Generate a working demo, traceable back to the stories
- Present a structured demo — and practise skilled prompting

1.5 days, full end-to-end

4 students per team

6–12 teams (24–60 students)

6 gated artifact stages

MIT's Aug 2026 report on AI in education warns of "cognitive surrender" when students accept AI output unchecked. This rubric's AI-collaboration score exists for exactly that reason — points for critiquing and correcting AI, not just using it.

NOT JUST ANOTHER HACKATHON

Traditional hackathons prove you can code. This one proves you can ship a product.

	TRADITIONAL HACKATHON	AI-PDLC HACKATHON
What it simulates	A dev sprint — idea to a working prototype	A full product team — market need to shipped demo
Roles you practice	Mostly Developer	Product/BA · Backlog/QA · Build · Demo — everyone rotates through all four
Core skill built	Speed & technical execution	Requirements thinking, structured collaboration & communication — plus code
How it's judged	Does the demo work?	MRD → PRD → backlog → working software — all traceable, then demoed
What's still true	Unmatched for technical depth, creative sprints & portfolio pieces	Builds the exact muscle employers say new graduates are missing

India's engineering graduates are ~70% technically "employable" — yet employers still cite communication and cross-functional collaboration as the single biggest hiring gap. A code-only hackathon doesn't touch that gap. This one is built to.

Sources: India Skills Report 2025–26; Unstop Talent Report 2025; MIT's Aug 2026 AI-in-education report — independently linking AI-driven isolation to exactly this kind of role-based teamwork, and naming project portfolios (not grades) as the assessment employers trust most.

PHASE 2 · SCALE

From one pilot to every department, every semester

1

Repeat the hackathon each semester

Same self-contained format; refresh the problem-statement bank per run.

2

Extend to every department — CS and non-CS alike

CSE, IT, AI/DS first; then Mech, Civil, Agri, Pharmacy and others with domain-specific problem sets, per NPAI's cross-sectoral model.

3

Embed AI awareness into general education

Foundational AI/GenAI modules for every student, not just participants — including non-technical disciplines.

4

Train faculty mentors as multipliers

2–4 mentors per run become the trainers for the next department's cohort.

PHASE 3 · INSTITUTIONALIZE

Make AI capability business-as-usual, not an annual event

1

Curriculum

Model curricula and Data & AI labs co-designed with industry; AI coursework credited into degree programmes across all departments.

2

Faculty

Train-the-Trainer AI academies certify faculty mentors each cycle. Each department names an AI Lead — the standing owner once the pilot's champion moves on, mirroring MIT's 2026 call for department-level ownership.

3

Incentives

Learning scholarships, research fellowships and innovation grants for students who go on to build. An AI Pilot Fund lets any department apply for credits, TA support or summer grants to run its own AI-aware experiments — the same model MIT adopted in 2026.

4

Industry & government

MoUs with tech employers for internships and hackathon judging; align with IndiaAI Mission and NPAI funding routes.

PHASE 3 · THE CULTURE LAYER

Curriculum and credits institutionalize the skill. Vibe institutionalizes the demand.

WHY THIS MATTERS

The pattern this is designed to avoid

Most co-curricular tech events follow the same arc: mandated onto a timetable or pushed through a class group, and tracked by attendance rather than energy. The same small group of student volunteers carries it each year, faculty spend more time chasing sign-ups than mentoring, and turnout quietly narrows until the event survives on the calendar but not in practice — then gets folded into a mandatory activity nobody looks forward to, or quietly dropped.

Event design is not a footnote to curriculum design — it decides whether students choose to be there, or whether the event is competing with an attendance sheet for their attention.

FIVE LEVERS WE'LL BORROW

1. **Curated, not mandatory**
   Students apply and get selected — scarcity creates pull, not a form everyone ignores.
2. **A signature moment**
   An unusual venue, a night format, a dramatic countdown — something worth telling friends about.
3. **Fest energy, not exam energy**
   Music, lighting, a live leaderboard on screen, a host running the room.
4. **Documentation that travels**
   Professional photo/video and an aftermovie — built for Instagram, not just the college newsletter.
5. **An identity of its own**
   Brand it, name it, keep it — spotlight past winners to build prestige year over year.

This is what turns institutionalization from a mandate into a magnet — students showing up because they want to, not because attendance is tracked.

LEARNING FROM OTHERS

Global benchmarks this roadmap draws on

Singapore	*AI for Everyone + SkillsFuture*	Segments talent into creators/practitioners/users; reimburses citizens for retraining.
United States	*AI Education Initiative*	Shared research infrastructure and an "AI Corps" to extend training and re-skill workers.
China	*AI Innovation Action Plan*	100 interdisciplinary AI majors, national online courses, dedicated AI schools.
Finland	*Elements of AI (AuroraAI)*	A free, public course pledging to educate 1% of the population in AI basics.

Source: Nasscom–Deloitte, "Advancing India's AI Skills" (Aug 2024)